

Clustering Algorithms

CS21206: Foundations of AI and ML

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Agenda

- § Understanding the concept of Clustering
- § Hierarchical Clustering
- § k-Means Clustering

Resources

- § Applied Machine Learning by David Forsyth.
- § The Elements of Statistical Learning by Trevor Hastie, Robert Tibshirani and Jerome Friedman.

Recall: Types of Learning



Supervised

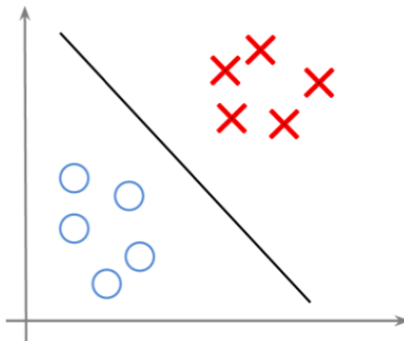


Unsupervised



Reinforcement

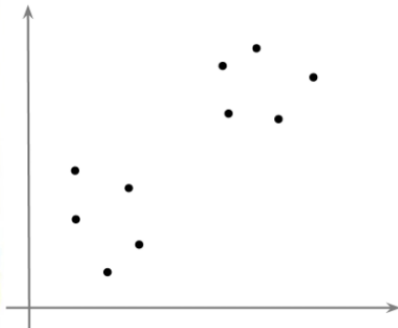
Supervised Learning



§ Training set: $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$

§ Example: Neural networks, SVM, Decision trees *etc.*

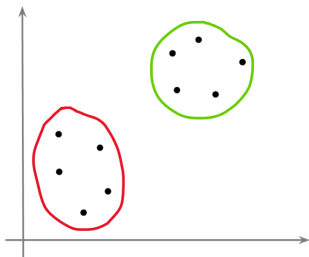
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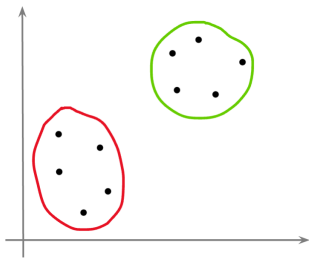
§ Example: PCA, Clustering Algorithms *etc.*

Clustering



- § One important task in unsupervised learning is finding hidden structures in data.
- § A hidden structure seems to be - the data points form two groups, called two clusters.
- § It is extremely useful and important to cluster data.

Clustering



- § One important task in unsupervised learning is finding hidden structures in data.
- § A hidden structure seems to be - the data points form two groups, called two clusters.
- § It is extremely useful and important to cluster data.
- § But it is surprisingly hard to define crisp criteria for a good (or bad) clustering.
- § Typically, one evaluates clustering by seeing how well it supports an application.

Agglomerative Clustering

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- § This can be somewhat harder than finding the distance between points.

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- § In **agglomerative clustering**, start with each data being a cluster and then merge clusters recursively.
- § Challenge is to know a good way to measure the distance between clusters.
- § This can be somewhat harder than finding the distance between points.
- § In **divisive clustering**, start with the entire dataset as a single cluster and split clusters recursively.
- § Challenge is to know some good criterion for splitting clusters.

Inter-cluster Distance

- § For agglomerative clustering, we need to choose a good inter-cluster distance to fuse nearby clusters. For example,
- § **Single-link clustering:** The distance between the closest elements of two clusters is chosen as the inter-cluster distance.
- § This tends to yield extended clusters.

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- § **Single-link clustering:** The distance between the closest elements of two clusters is chosen as the inter-cluster distance.
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- § **Complete-link clustering:** The maximum distance between an element of the first cluster and one of the second is chosen as the inter-cluster distance.
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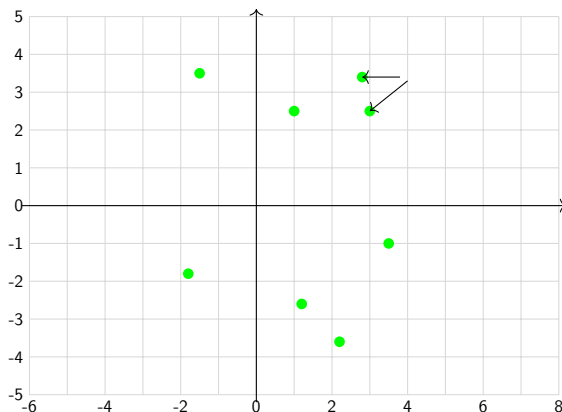
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- § **Complete-link clustering:** The maximum distance between an element of the first cluster and one of the second is chosen as the inter-cluster distance.
- § This tends to yield rounded clusters.
- § **Group average clustering:** The average of distances between all pairs of elements in the clusters is chosen as the inter-cluster distance.
- § This also tends to yield rounded clusters.

Single-link clustering

Agglomerative Clustering

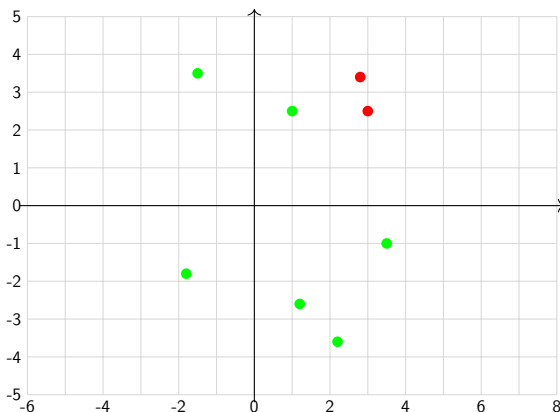
Make each point a separate cluster. Until the clustering is satisfactory, merge the two clusters with the smallest inter-cluster distance.



Single-link clustering

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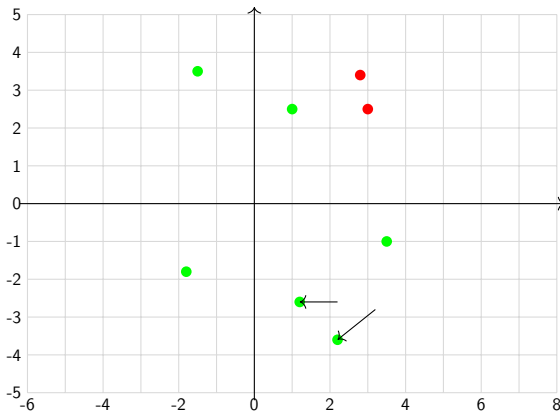
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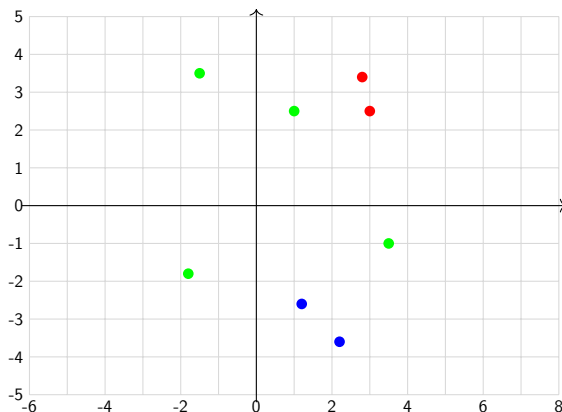
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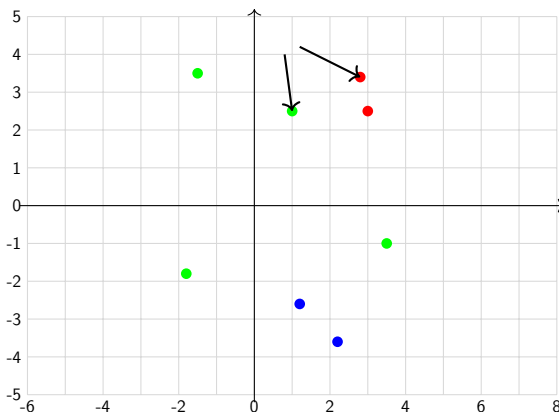
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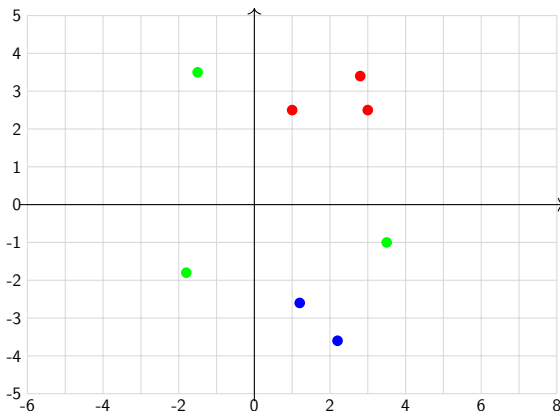
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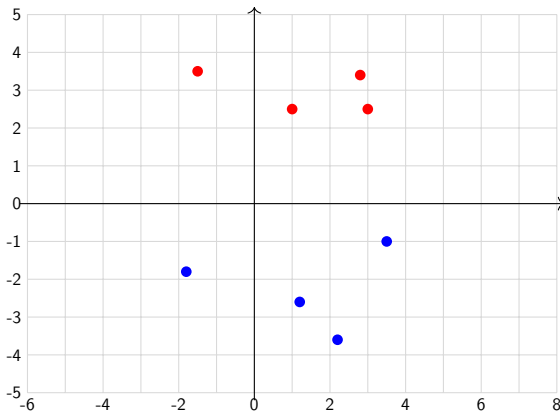
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- § Agglomerative clustering generates a hierarchy of clusters represented as a **dendrogram**
- § An appropriate choice of clusters is made from the dendrogram.

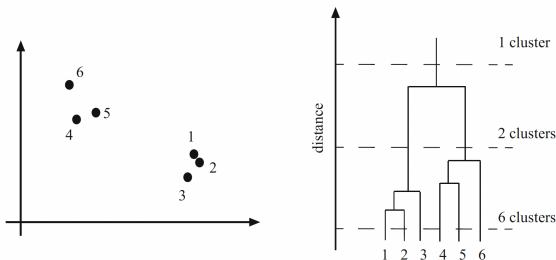


Figure 8.1: **Left**, a dataset; **right**, a dendrogram obtained by agglomerative clustering using single-link clustering. If one selects a particular value of distance, then a horizontal line at that distance splits the dendrogram into clusters. This representation makes it possible to guess how many clusters there are and to get some insight into how good the clusters are

Fig. credit: *Applied Machine Learning* by D. Forsyth

K-Means Clustering

- § One popular group of clustering algorithms directly assign each observation to a cluster.
- § Each data is uniquely labeled by an integer $i \in \{1, 2, \dots, N\}$.
- § A prespecified number of clusters $k < N$ is postulated and each data is assigned to one and only one cluster.
- § These assignments can be characterized by an *encoder* $k = C(i)$, that assigns the i^{th} observation to the k^{th} cluster.
- § One seeks the particular encoder $C^*(i)$ that achieves the required goal, based on the dissimilarities $d(\mathbf{x}_i, \mathbf{x}_{i'})$ between every pair of data.

Direct Loss Function Approach

Since the goal is to group close points, a natural loss function is the **Within-Cluster** Point Scatter:

Within-Cluster Point Scatter $W(C)$

$$W(C) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \sum_{C(i')=k} d(\mathbf{x}_i, \mathbf{x}_{i'})$$

- § where, $d(\mathbf{x}_i, \mathbf{x}_{i'})$ is the distance between points $\mathbf{x}_i$ and $\mathbf{x}_{i'}$ (e.g., squared Euclidean).
- § $W(C)$: gives how close data points are within the same cluster.

Total Scatter Decomposition

The **Total Point Scatter** T is given by,

$$\begin{aligned} T &= \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \left(\sum_{C(i')=k} d_{i,i'} + \sum_{C(i') \neq k} d_{i,i'} \right) \\ &= W(C) + B(C) \end{aligned}$$

Where $d_{i,i'} = d(\mathbf{x}_i, \mathbf{x}_{i'})$ and **Between-Cluster Point Scatter** $B(C)$ is:
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$$B(C) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \sum_{C(i') \neq k} d_{i,i'}$$

Key Insights:

§ T is constant given the data.

§ Since $W(C) = T - B(C)$, **minimizing within-cluster scatter** is mathematically equivalent to **maximizing between-cluster scatter**.

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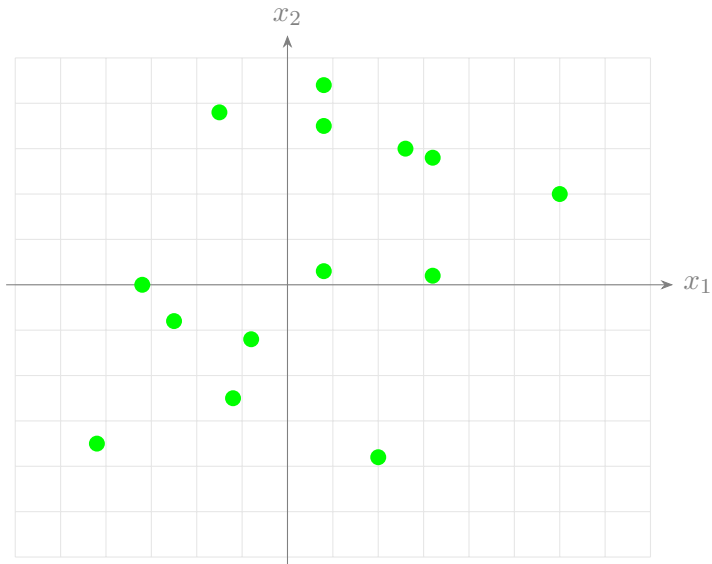
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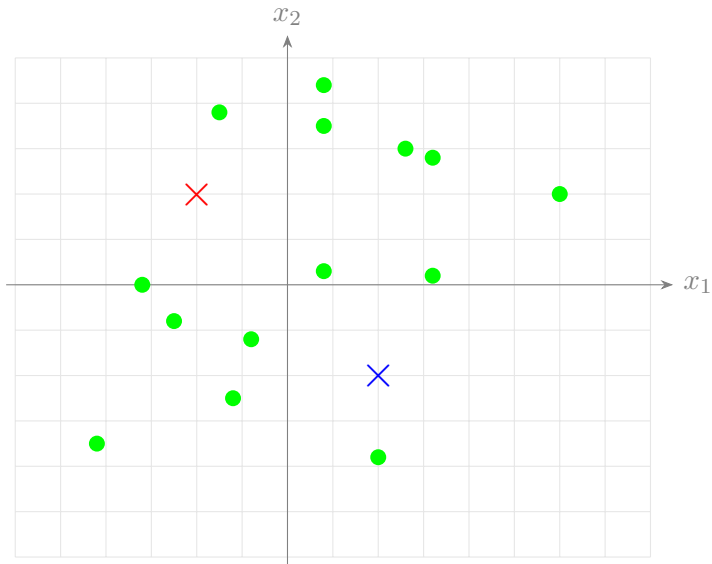
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- § For $N = 19$ and $K = 4$, there are almost 10^{10} possible partitions.
- § This makes exhaustive search impossible (NP-Hard). We rely on iterative heuristics like **Lloyd's Algorithm**.

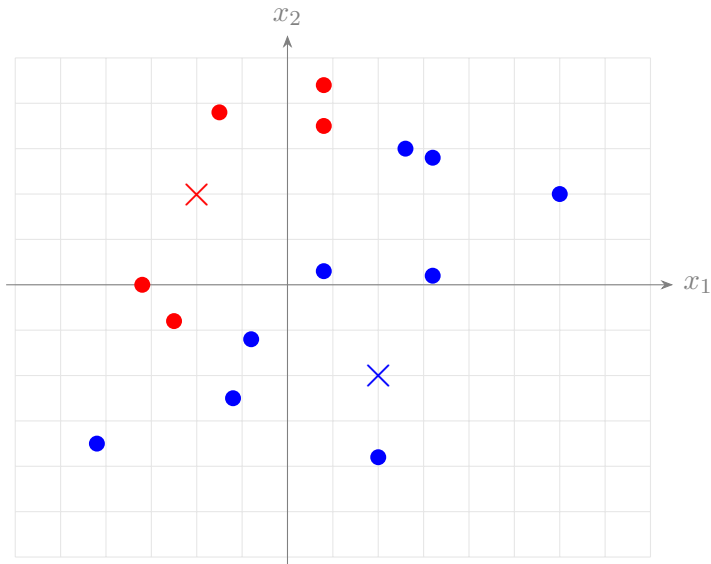
K-Means Iteration Visualized



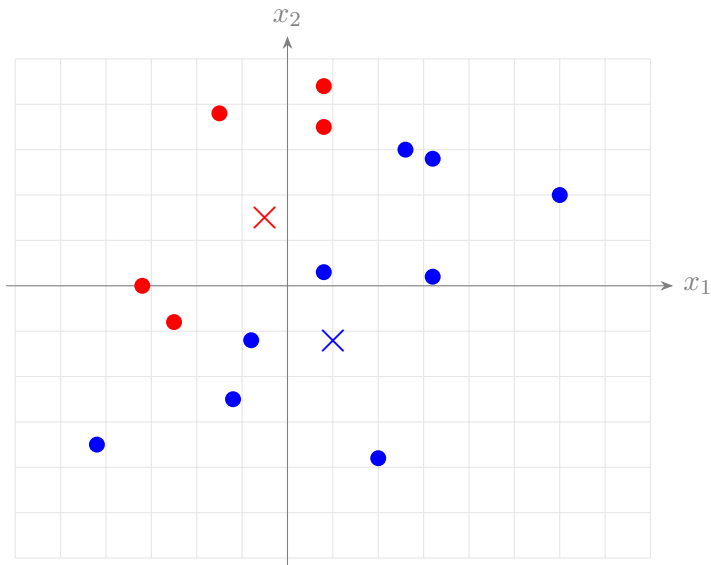
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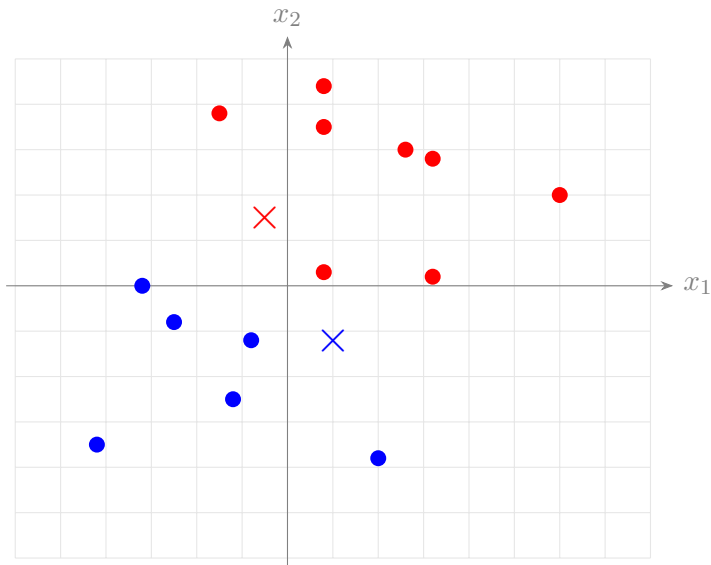
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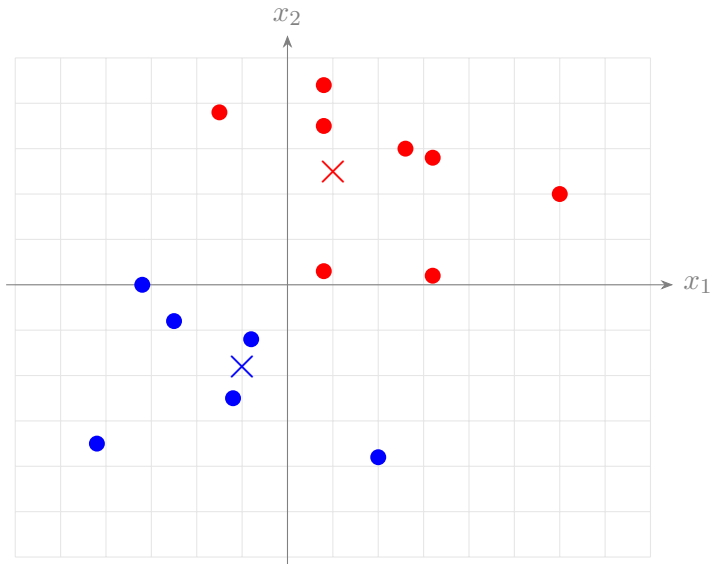
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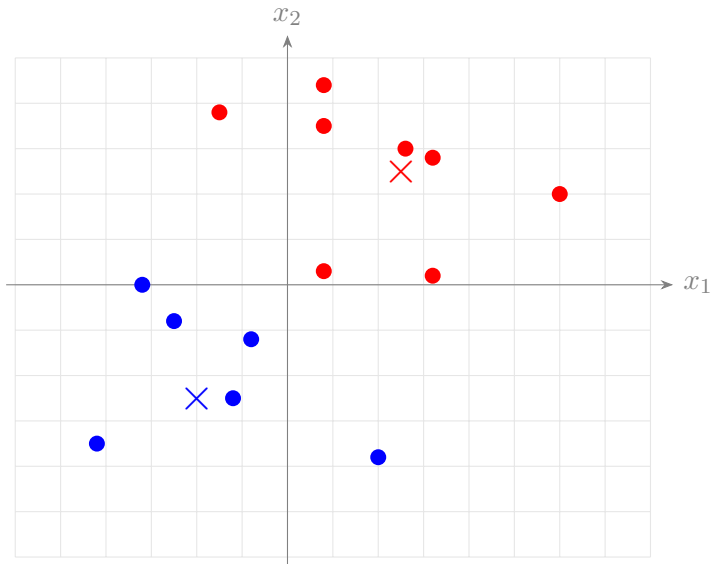
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K-Means as Approximate Solution

- § The algorithm we witnessed above is a remarkably effective approximate solution to the combinatorial problem.
- § The main strength is that the assignment of datapoints to clusters at any iteration is a perturbation of that for the previous iteration, so only a very small fraction of all possible assignments are examined.

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- § Coming back to the loss function we derived earlier,

$$W(C) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \sum_{C(i')=k} d(\mathbf{x}_i, \mathbf{x}_{i'}) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \sum_{C(i')=k} \|\mathbf{x}_i - \mathbf{x}_{i'}\|^2 \quad (1)$$

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- § This can further be written as,

$$W(C) = \sum_{k=1}^K N_k \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad (2)$$

where $\boldsymbol{\mu}_k$ is the mean of the mean of the datapoints associated with the k^{th} cluster and N_k is the number of such datapoints.

Deriving the Within-Cluster Objective (Centroid View)

We start with the pairwise term:

$$\begin{aligned}\|\mathbf{x}_i - \mathbf{x}_{i'}\|^2 &= \|(\mathbf{x}_i - \boldsymbol{\mu}_k) - (\mathbf{x}_{i'} - \boldsymbol{\mu}_k)\|^2 \\ &= [(\mathbf{x}_i - \boldsymbol{\mu}_k) - (\mathbf{x}_{i'} - \boldsymbol{\mu}_k)]^T [(\mathbf{x}_i - \boldsymbol{\mu}_k) - (\mathbf{x}_{i'} - \boldsymbol{\mu}_k)] \\ &= \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 + \|\mathbf{x}_{i'} - \boldsymbol{\mu}_k\|^2 - 2(\mathbf{x}_i - \boldsymbol{\mu}_k)^T (\mathbf{x}_{i'} - \boldsymbol{\mu}_k) \quad (3)\end{aligned}$$

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 \end{aligned}$$

Now, we sum over all i and i' in the cluster:

$$\begin{aligned}
 \sum_{C(i)=k} \sum_{C(i')=k} \|\mathbf{x}_i - \mathbf{x}_{i'}\|^2 &= \sum_{C(i)=k} \sum_{C(i')=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 + \|\mathbf{x}_{i'} - \boldsymbol{\mu}_k\|^2 - 2(\mathbf{x}_i - \boldsymbol{\mu}_k)^T (\mathbf{x}_{i'} - \boldsymbol{\mu}_k) \\
 &= N_k \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 + N_k \sum_{C(i')=k} \|\mathbf{x}_{i'} - \boldsymbol{\mu}_k\|^2 \\
 &\quad - 2 \underbrace{\sum_{C(i)=k} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T}_{0} \underbrace{\sum_{C(i')=k} (\mathbf{x}_{i'} - \boldsymbol{\mu}_k)}_{0} \quad (4)
 \end{aligned}$$

By definition of the mean, the cross term vanishes.

Deriving the Within-Cluster Objective (Centroid View)

$$\sum_{C(i)=k} \sum_{C(i')=k} \|\mathbf{x}_i - \mathbf{x}_{i'}\|^2 = 2N_k \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad (5)$$

Putting this in eqn. 1 gives eqn. 2.

$$W(C) = \sum_{k=1}^K N_k \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad (6)$$

§ So, what minimizes $W(C)$?

§ If each term of the outer summation is minimized.

§ Note that - $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$ is already in minimized form. What we mean by this is: - for any set of observations S , the point m that minimizes the sum of squared distances is the mean $\boldsymbol{\mu}_S$:

$$\boldsymbol{\mu}_S = \arg \min_m \sum_{i \in S} \|\mathbf{x}_i - m\|^2$$

§ **Proof Sketch:** Take the derivative of $f(m) = \sum_{i \in S} \|\mathbf{x}_i - m\|^2$ with respect to m and equate it to 0

The Enlarged Optimization Problem

By treating the centers $\{\mu_k\}_1^K$ as variables, we get the **K-means enlarged problem**:

Enlarged Optimization Problem

$$\min_{C, \{\mu_k\}_1^K} \sum_{k=1}^K N_k \sum_{C(i)=k} \|x_i - \mu_k\|^2$$

This allows for an **alternating optimization** procedure:

- 1 **Fix Centers, Optimize Partition:** Assign each x_i to the closest μ_k .
- 2 **Fix Partition, Optimize Centers:** Set μ_k = the new mean of points in cluster k .

This is exactly Lloyd's Algorithm!

Strategies for Initialization

Initialization significantly impacts the quality of the final clustering result.

- § **Random Initialization:** A natural strategy is to choose k data items at random and use them as initial cluster centers.
- § **Sensitivity:** K-means is sensitive to initialization; an "unlucky" choice of initial points can lead to poor local optima.
- § **Multiple Restarts:** A widely adopted strategy is to run the algorithm several times with different initializations and choose the best result.
- § **K-means++:** A more sophisticated approach with theoretical guarantees.
- § **K-means++ Procedure:** Choose the first center $\mathbf{x}$ uniformly at random.
- § **Probability Distribution:** Choose subsequent $k - 1$ centers as IID draws from:

$$P(\text{choose } i\text{-th point}) = \frac{d_i^2(\mathbf{x})}{\sum_u d_u^2(\mathbf{x})}$$

How to Choose K ?

Usually, we do not know how many clusters there should be and must determine K through experimentation.

- § **Cost Function Behavior:** We expect the cost function value to go down as K goes up, as points find centers closer to them.
- § **The "Knee" Method:** It is helpful to plot the cost function value as a function of K and look for the "knee" (or elbow) of the curve.
- § **Reasonable Choices:** Multiple values (e.g., $K = 2, 3$, or 4) might all seem like reasonable choices based on the plot.
- § **Purity:** In special cases with known ground truth, clustering quality can be evaluated by the **purity** (number of different labels in a cluster).
- § **Downstream Metrics:** Often, the best K is chosen based on how well the clusters perform in a specific later application.